

## No. of Functional Features Included in the Solution

|                      |               |
|----------------------|---------------|
| <b>Date</b>          | 30 June 2026  |
| <b>Team ID</b>       |               |
| <b>Project Name</b>  | Rising Waters |
| <b>Maximum Marks</b> | 5 Marks       |

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

### Functional Features Overview

| # | Feature Name                   | Description                                                  | Module             | Status | Marks |
|---|--------------------------------|--------------------------------------------------------------|--------------------|--------|-------|
| 1 | Workspace Slider Interface     | Allows adjusting GNI, Health, and Schooling variables        | UI Dashboard       | Done   | 1.5   |
| 2 | AJAX Prediction API            | Computes predicted HDI values on the fly without page reload | Backend logic      | Done   | 1.5   |
| 3 | Country Baseline Preset Loader | Loads official UNDP country baselines on request             | Backend API        | Done   | 0.5   |
| 4 | Relational Simulation Logger   | Saves simulation input parameters and score outputs          | Database Storage   | Done   | 0.5   |
| 5 | Dynamic SVG Report Generator   | Generates vector graphics showing development tiers          | Backend Controller | Done   | 0.5   |
| 6 | User Authentication Service    | Secures analyst login and dashboard sessions                 | Security           | Done   | 0.5   |

### Feature Summary

| Metric                             | Count / Value |
|------------------------------------|---------------|
| Total Features Planned             | 6             |
| Total Features Implemented         | 6             |
| Core / Must-Have Features          | 6             |
| Additional / Nice-to-Have Features | 0             |
| Features Tested & Verified         | 6             |

**Feature Category Breakdown**

| # | Category                  | Features in Category           | Example Features               |
|---|---------------------------|--------------------------------|--------------------------------|
| 1 | User Interface (UI)       | Sliders, Country Templates     | Interactive workspace inputs   |
| 2 | Backend / Logic           | Flask App, Model Inference     | Dynamic forecast API           |
| 3 | Database / Storage        | SQLite, SQLAlchemy ORM         | Session and prediction history |
| 4 | API / Integration         | AJAX REST endpoints            | /api/predict, /api/countries   |
| 5 | Security / Authentication | User Login, Session validation | Auth logic gates               |